

# Perfect Competition

# Types of markets

Different firms operate in different markets/industries with different degrees of competition. Generally, we have four major types of markets:

- *Perfect competition: large number of firms selling identical products*
- Monopolistic competition: large number of firms but with some degree of product differentiation
- Oligopoly: small number of firms with some degree of product differentiation
- Monopoly: One firm producing a good/service which has no close substitutes

# A competitive market

- This chapter analyzes how firms make production decisions in a competitive market.
- The characteristics of a *competitive market* are:
  - *Many* firms sell *identical* products to *many* buyers.
  - Buyers and sellers are *price takers*.
  - *No barriers* to enter and exit the market.
  - Everyone has *perfect information* regarding prices.

# Perfect competition

- If a buyer or seller has the ability to noticeably affect market prices, that person/firm has *market power*.
- Competitive markets have so much competition that *no one* has the ability to affect market prices. Thus, all are *price takers*.
- *Demand* for an individual firm's products (not the same as market demand for the good) is *perfectly elastic* – since *perfect substitutes exist*, people will switch to other firms
- If you sell higher than the market price – lose all market share
- If you sell lower than the market price – giving away revenue

# Perfect competition- extras

Why is competition good?

- Lower prices
- Higher quality goods and services
- More innovation (to produce most efficiently)
- Unproductive firms are driven out of business
- Creates level-playing field for small businesses

# Explicit and implicit costs - extras

- A firm's opportunity cost of operation has two components.
- The first is composed of the fixed and variable costs.
  - These are *explicit costs* that require a firm to spend money.
- The second is composed of forgone opportunities.
  - These are *implicit costs* that represent opportunities that could have generated revenue if the firm had invested its resources in another way.
- To properly account for the total costs incurred by a firm, total cost includes both types of costs.

# Economic and accounting profit - extras

- When companies report their profits, they provide *accounting profits*:

Accounting profit = Total revenue – Explicit costs

- Accounting profits may be a misleading indicator of how well a business is really doing.
- To account for implicit costs, *economic profit* further subtracts implicit costs:

Economic profit = Accounting profit – Implicit costs

# Profit Maximization

Firm's Goal: Profit Maximization

Recall

- *Profit* = Total Revenue – Total Cost
- *Total Revenue* = Price X Quantity
- Marginal Revenue: Change in total revenue that results from a one unit increase in quantity sold

$$\textit{Marginal Revenue} = \Delta \textit{Total Revenue} / \Delta \textit{Quantity}$$



# Profit Maximization

- Firms seek to maximize profits - they need to increase their total revenue ( $P \times Q$ )
- In a competitive market, the only choice that a *price-taking* firm can make to affect profits is the *quantity of output* to produce.
- The profit-maximizing quantity corresponds to the quantity at which *marginal revenue is equal to the marginal cost*.

## Example

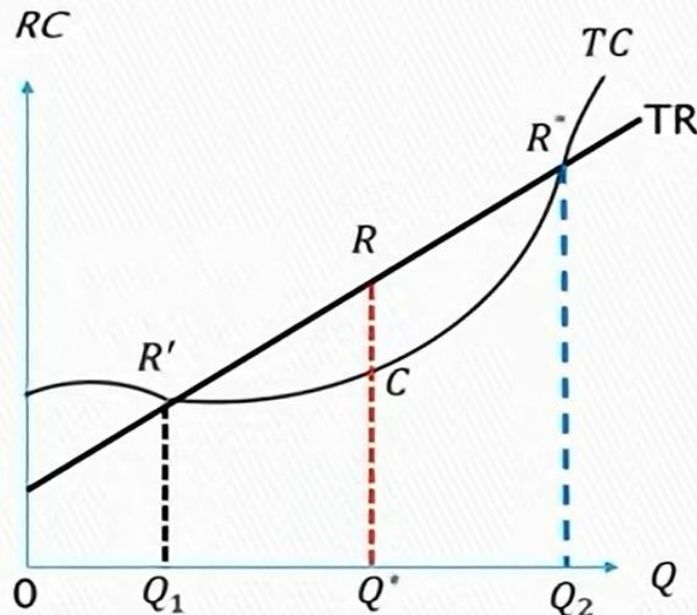
A milk producing firm under perfect competition sells gallons of milk (various quantities) at a market price,  $P=6$  per gallon. Calculate the total, average and revenue of this firm.

| Quantity<br>(Q) | Price<br>(P) | Total<br>Revenue<br>(TR) | Average Revenue<br>$AR = \frac{\text{Total Revenue (TR)}}{\text{Quantity (Q)}}$ | Marginal Revenue<br>$MR = \frac{\text{New(TR)} - \text{last (TR)}}{\text{New(Q)} - \text{Last (Q)}}$ |
|-----------------|--------------|--------------------------|---------------------------------------------------------------------------------|------------------------------------------------------------------------------------------------------|
| 1               | 6            | 6                        | 6                                                                               | –                                                                                                    |
| 2               | 6            | 12                       | 6                                                                               | 6                                                                                                    |
| 3               | 6            | 18                       | 6                                                                               | 6                                                                                                    |
| 4               | 6            | 24                       | 6                                                                               | 6                                                                                                    |
| 5               | 6            | 30                       | 6                                                                               | 6                                                                                                    |
| 6               | 6            | 36                       | 6                                                                               | 6                                                                                                    |
| 7               | 6            | 42                       | 6                                                                               | 6                                                                                                    |
| 8               | 6            | 48                       | 6                                                                               | 6                                                                                                    |

**At,  $Q=8$ , where,  $TR = 48$  (highest), is the profit maximized?**

### *Total Approach*

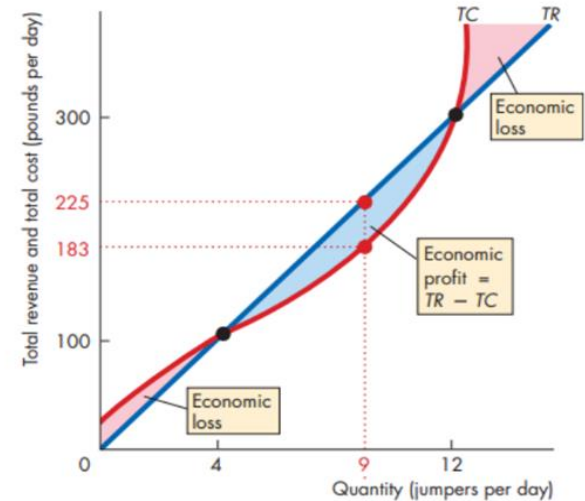
- ▶ Profit maximizes at the point where the difference between total revenue (TR) and total cost (TC) is the largest. This point also provides the firm with profit maximizing level of output.
- ▶ In this graph the difference between  $Q^*R$  and  $Q^*C$  is the highest, representing the maximum profit earned by the firm under total approach concept.
- ▶  $Q^*$  is the profit maximizing level of output.



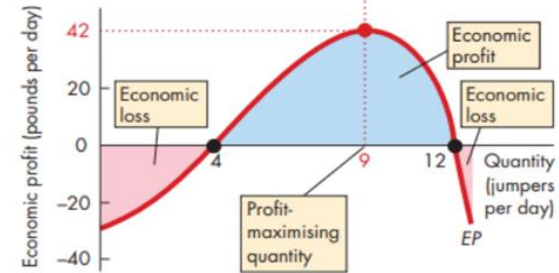
# Perfect Competition

## Firm's Output Decision

- Part a graphs TR & TC against Quantity produced
- Vertical distance between TR and TC gives profits
- Part b graphs Economic Profits against Quantity
- At low levels of production, firm is making losses – its production level is not high enough to cover its fixed costs
- At output levels greater than 4 jumpers and less than 12 jumpers, the firm is enjoying varying levels of economic profit, with profits reaching a peak at 9 jumpers per day
- At greater levels of output, firm's costs are steeply rising (due to diminishing returns to capital/labor) and therefore firm is making losses once again



(a) Revenue and cost



(b) Economic profit and loss

# Marginal Approach

Another way to find the profit-maximizing level of output is to use Marginal Approach

- Compare MR with MC
- In perfectly competitive markets, MR which equals price remains constant
- However, with increasing output, MC changes
- At low output levels, marginal cost decreases as output increases, but eventually marginal cost increases (law of diminishing marginal returns)

*Economic profits are maximized when  $MR=MC$*

## Marginal Approach

Here, the profit maximizing quantity is 5 where,  $MR = MC$ .

**Profit maximization condition;  $MR = MC$  or,  $(MR - MC = 0)$ . At this point optimal amount of output is produced. (Values are in \$ terms).**

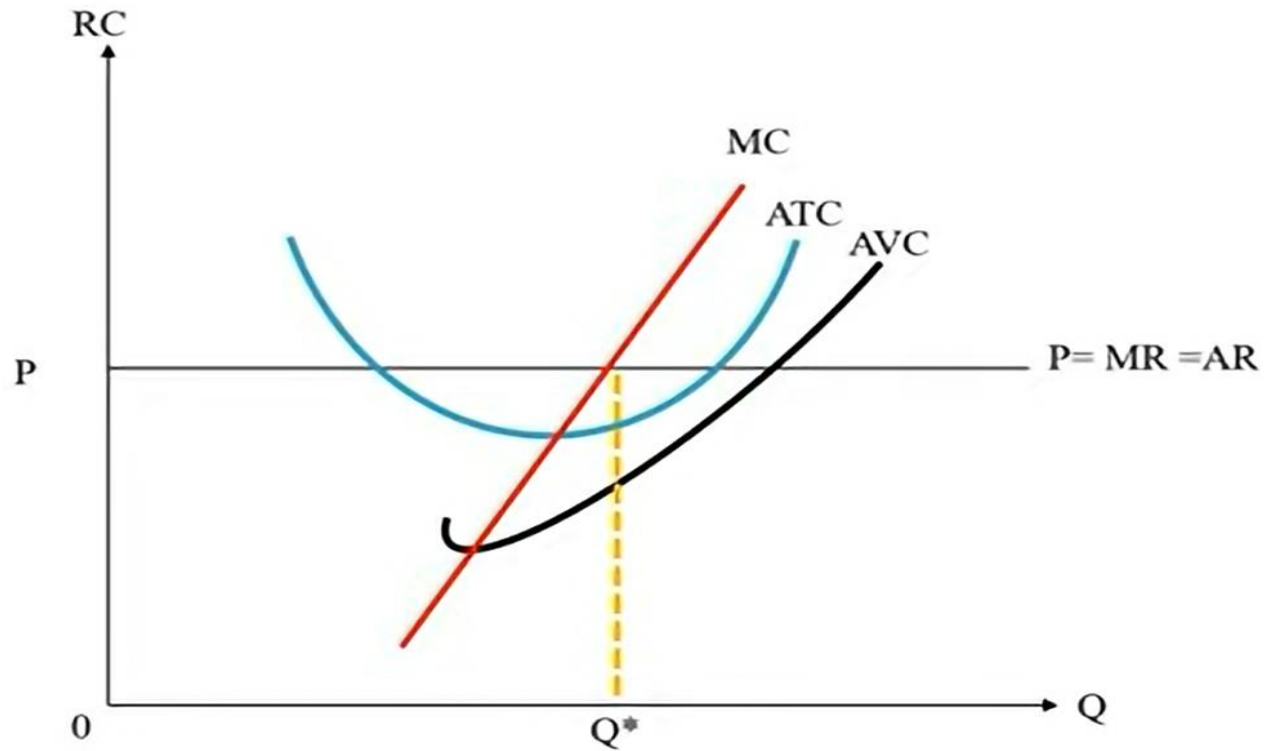
| Quantity (Q) | Total Revenue (TR) | Total cost (TC) | Profit (TR-TC) | Marginal Revenue $MR = \Delta TR / \Delta Q$ | Marginal Cost $MC = \Delta TC / \Delta Q$ | MR - MC |
|--------------|--------------------|-----------------|----------------|----------------------------------------------|-------------------------------------------|---------|
| 0            | 0                  | 3               | -3             | -                                            | -                                         | -       |
| 1            | 6                  | 5               | 1              | 6                                            | 2                                         | 4       |
| 2            | 12                 | 8               | 4              | 6                                            | 3                                         | 3       |
| 3            | 18                 | 12              | 6              | 6                                            | 4                                         | 2       |
| 4            | 24                 | 17              | 7              | 6                                            | 5                                         | 1       |
| 5            | 30                 | 23              | 7              | 6                                            | 6                                         | 0       |
| 6            | 36                 | 30              | 6              | 6                                            | 7                                         | -1      |
| 7            | 42                 | 38              | 4              | 6                                            | 8                                         | -2      |
| 8            | 48                 | 47              | 1              | 6                                            | 9                                         | -3      |



# Perfect Competition: Graphical Representation

- We know, in a competitive firm:  $P = AR = MR$
- Under perfect competition market, as both buyers and sellers are price takers; price is represented by a horizontal line as price is same regardless of the quantity purchased or sold. Hence, the *demand curve* is *perfectly elastic* under perfect competition.
- *MC curve* is upward sloping and works as a *supply curve* under perfect competition; as marginal cost of a firm determines the quantity of good the firm is willing to supply at a given price.

# Profit Maximization of a Competitive Firm

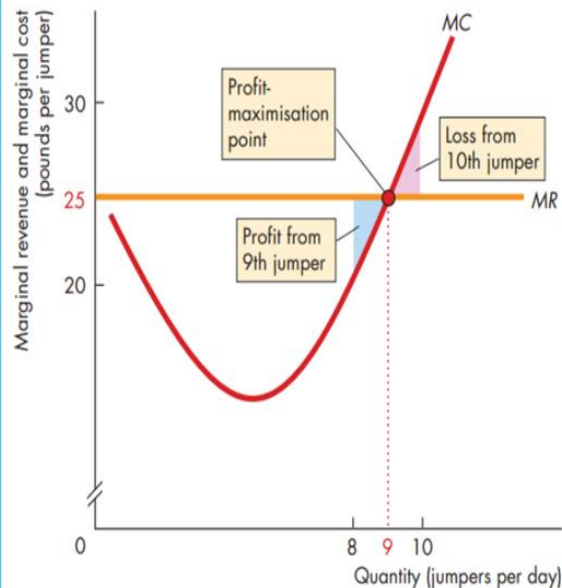




## Firm's Output Decision

- If marginal revenue exceeds marginal cost (if  $MR > MC$ ), then the extra revenue from selling one more unit exceeds the extra cost incurred to produce it.
- The firm makes an economic profit on the marginal unit, so economic profit increases if output increases.
- If marginal revenue is less than marginal cost (if  $MR < MC$ ), then the extra revenue from selling one more unit is less than the extra cost incurred to produce it.
- The firm incurs an economic loss on the marginal unit, so its economic profit decreases if output increases
- If  $MR = MC$ , economic profit decreases if output changes in either direction, so economic profit is maximized.

**Figure 11.3** Profit-Maximising Output



# Perfect Competition: Output decisions

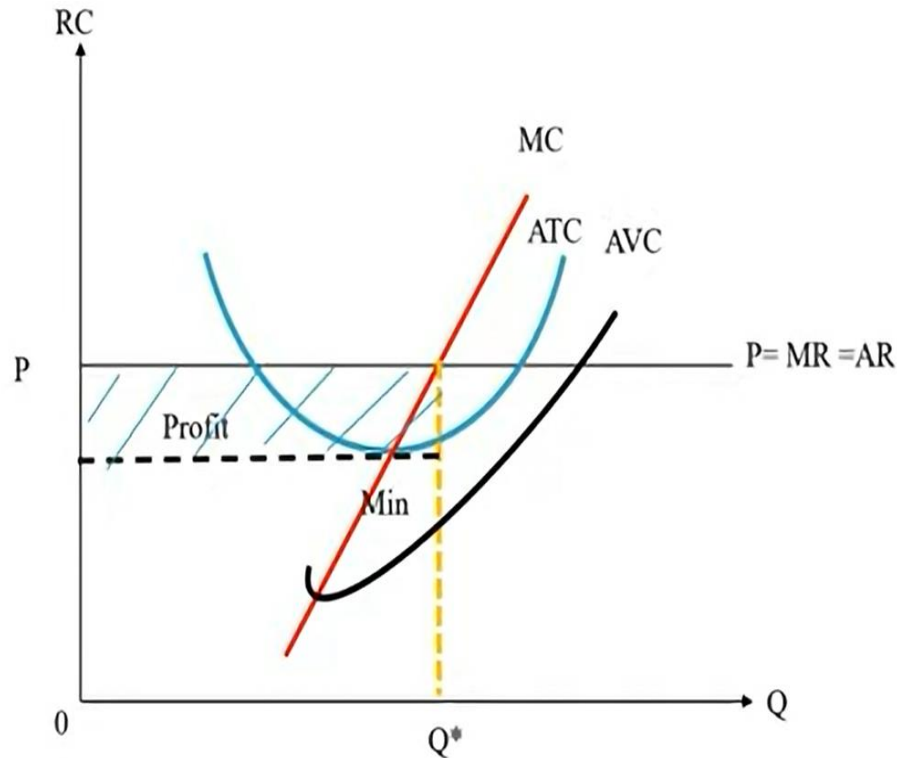
- When  $MR > MC$ , the firm should keep on producing as it will generate profit for the firm
- When  $MR < MC$ , the firm should not produce anymore as they will incur loss from any further level of production
- When  $MR = MC$ , the firm ensures the highest level of profit and provides the optimum level of output at which profit of the firm is maximized.

# Price and Cost comparison: Output decisions

- Market price is a deciding factor for the firms whether to enter or exit the market.
- Depending on the price and cost, the firms will decide whether they will continue their production, shutdown temporarily in the short-run or stay/exit the market in the long-run.
- Comparing between price and ATC, AFC and AVC helps the firm to decide their output levels in the *short-run and long-run*.

# Profit / Abnormal Profit Under Perfect Competition

$$P > \min ATC$$

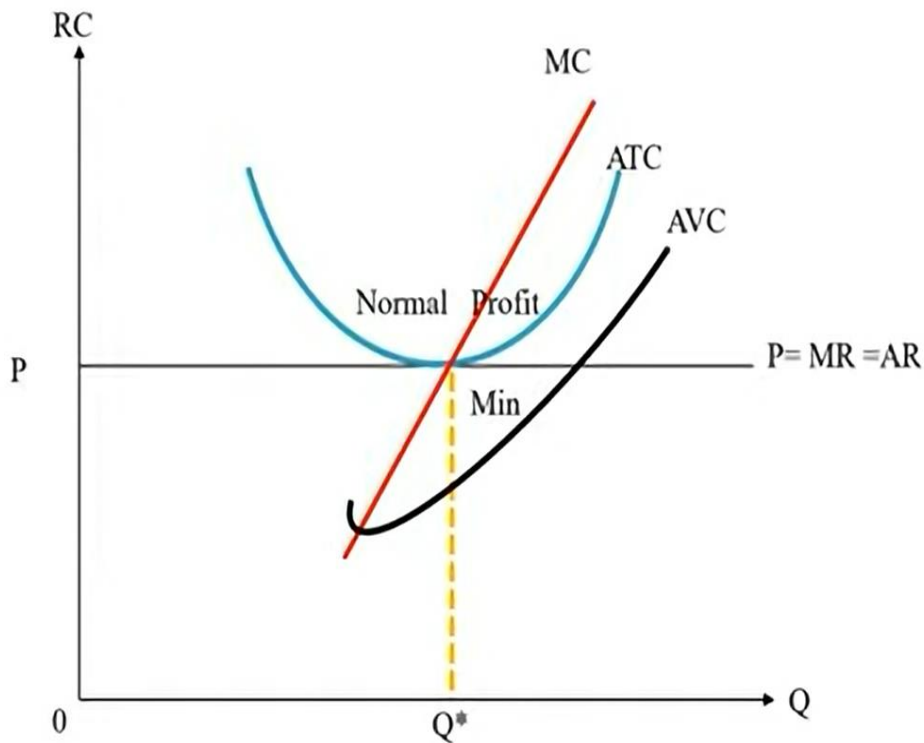


If *Price* > *min ATC*, the firm is making *abnormal profit*. They will continue production in the short-run to remain in the business and will also continue in the long-run.

If *Price = min ATC*, then the firm is covering both of its AFC and AVC. In that case, the firm is making *normal profit*. This is also known as the *Breakeven point* under perfect competition. The firm will continue its production in the short-run. In the long-run, the firm is indifferent between staying in operation or shutting down.

## Normal Profit Under Perfect Competition

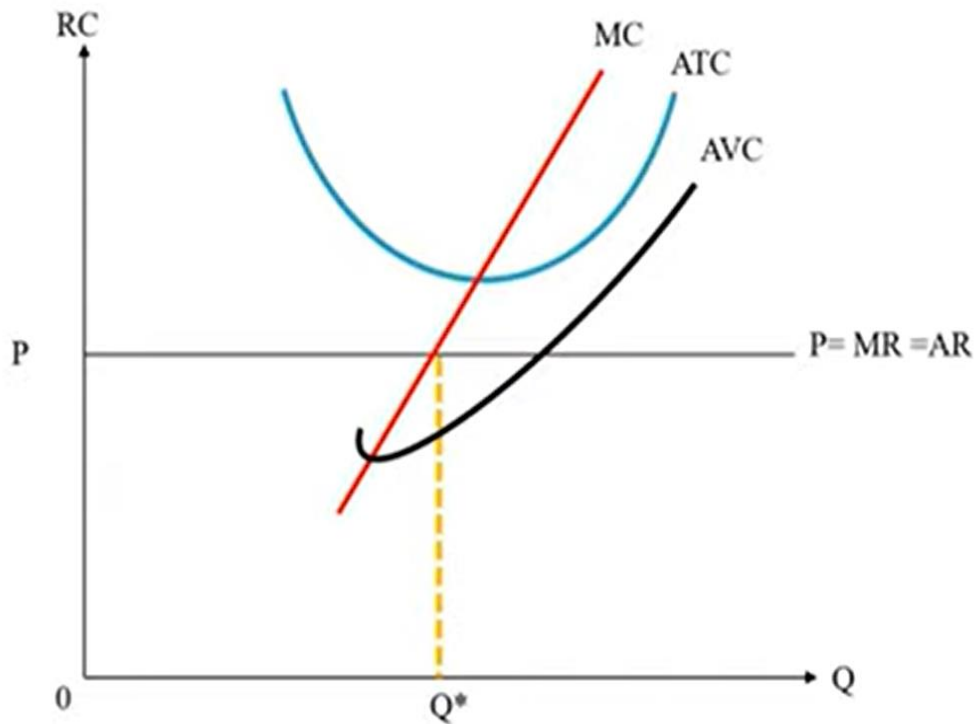
### $P = \min ATC$ (Breakeven Point)



## Under Perfect Competition $\min AVC < P < \min ATC$

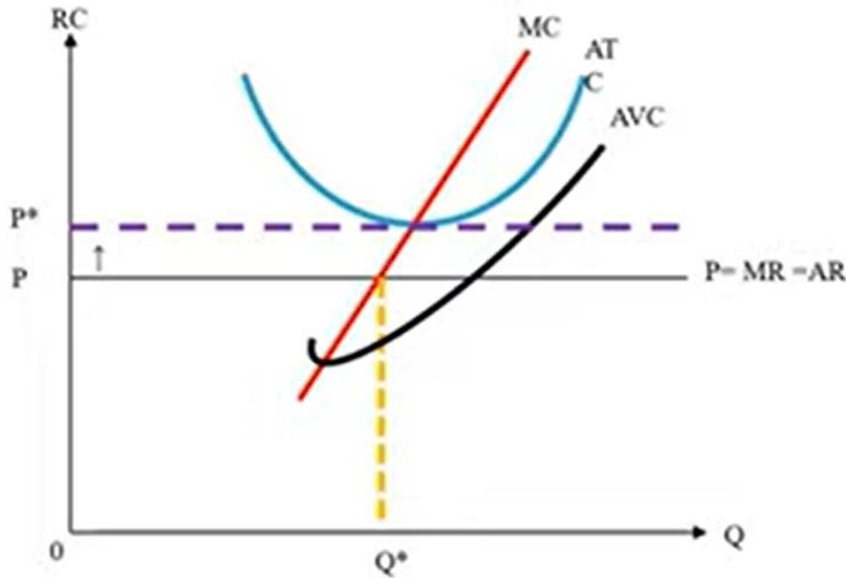
If *Price*  $< \min ATC$ , but *Price*  $> \min AVC$ , then the firm cannot cover all of its AVC, as a result, it will incur *loss*.

The firm might continue production in the short-run to cover its fixed costs and some of the variable costs and eventually exit the market in the long-run.



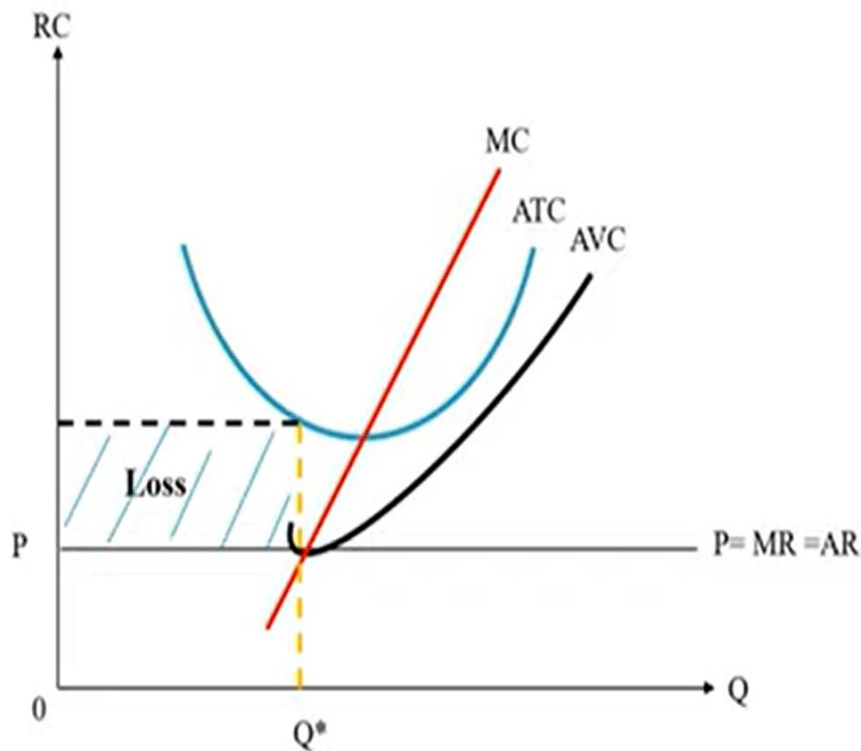
## In long run if price increases

The firm might continue production in the long-run if the *price level increases* and they are able to cover up for the loss. Whether the firm will continue or will stop production to save the variable cost, depends on the firm's decision.



## Loss incurred by firms under perfect competition

$$P = \min AVC$$



If *Price = min AVC*, is regarded as the *shutdown point* under perfect competition. At this point, firms should not go for any new production. Rather, they just continue to cover up its fixed costs and need to wait for the price level to increase so that they can continue in the long-run.



# Price and AVC comparison

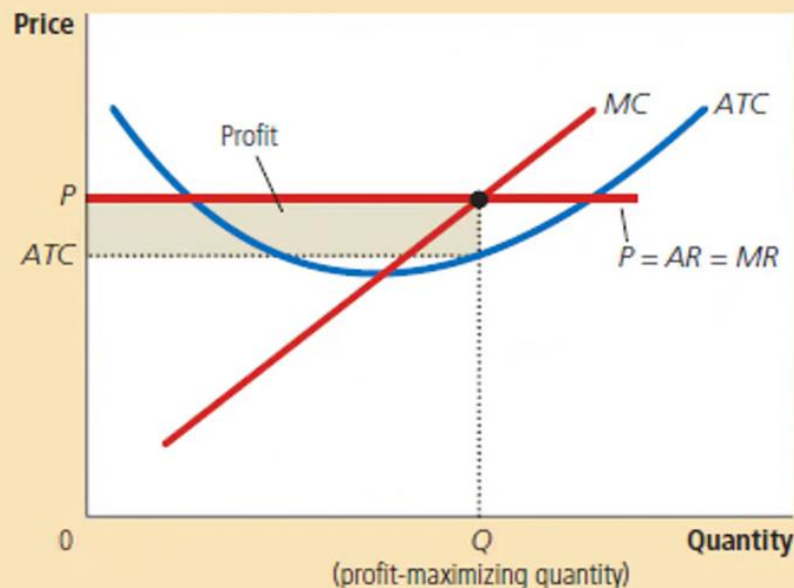
- If *Price* < *min AVC*, then firms cannot even cover its fixed costs. Hence, they cannot continue their business anymore and needs to shut down immediately.
- If *Price* > *min AVC* in the long-run, then the firm can reopen its business in the future.
- If it does not reopen in the long-run, the firm leaves or exits the market permanently.

## Figure 5

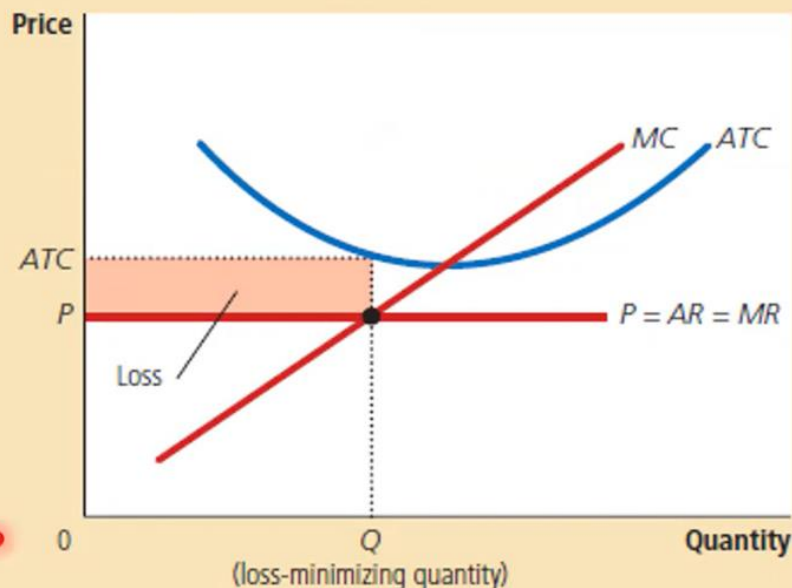
### Profit as the Area between Price and Average Total Cost

The area of the shaded box between price and average total cost represents the firm's profit. The height of this box is price minus average total cost ( $P - ATC$ ), and the width of the box is the quantity of output ( $Q$ ). In panel (a), price is above average total cost, so the firm has positive profit. In panel (b), price is less than average total cost, so the firm has losses.

(a) A Firm with Profits



(b) A Firm with Losses



| Quantity | Total Cost | Average Total Cost | Variable Cost | Average Variable Cost | Marginal Cost |
|----------|------------|--------------------|---------------|-----------------------|---------------|
| 0        | 2          | Undefined          | 0             | 0                     |               |
| 1        | 7          | 7                  | 5             | 5                     | 5             |
| 2        | 10         | 5                  | 8             | 4                     | 3             |
| 3        | 15         | 5                  | 13            | 4.3                   | 5             |
| 4        | 22         | 5.5                | 20            | 5.5                   | 7             |
| 5        | 31         | 6.2                | 29            | 5.8                   | 9             |
| 6        | 42         | 7                  | 40            | 6.6                   | 11            |

Complete the table above

1. At what level of average total cost does the firm reach its break-even point? Tk 5
2. At what level of average variable cost does the firm reach its shutdown point? TK 4

## Example

| Q  | FC  | VC   | TC   | AF<br>C | AVC                           | ATC                           | MR | MC | MR-MC |
|----|-----|------|------|---------|-------------------------------|-------------------------------|----|----|-------|
| 0  | 100 | 0    | 100  | -       | 0                             | 0                             | 21 | -  | -     |
| 10 | 100 | 200  | 300  | 10      | 20                            | 30                            | 21 | 20 | 1     |
| 20 | 100 | 300  | 400  | 5       | 15                            | 20                            | 21 | 10 | 11    |
|    |     |      |      |         | Min AVC<br>Shut down<br>point |                               |    |    |       |
| 30 | 100 | 480  | 580  | 3.33    | 16                            | 19.33                         | 21 | 18 | 3     |
|    |     |      |      |         |                               | Min ATC<br>Breakeven<br>point |    |    |       |
| 40 | 100 | 700  | 800  | 2.5     | 17.5                          | 20                            | 21 | 22 | - 1   |
| 50 | 100 | 1000 | 1100 | 2       | 20                            | 22                            | 21 | 30 | - 9   |

## Example

- ▶ Price = MR = \$ 21
- ▶  $TR = 21 \times 30 = 630$  (at the profit maximizing level)
- ▶ Break even price = 19.33 (Normal Profit)
- ▶ Shut down price = \$15
- ▶ Profit =  $TR - TC = 630 - 580 = \$ 50$
- ▶ As  $P > \min ATC$ , the firm earns abnormal profit. Can cover both AFC and AVC.
- ▶ If price becomes 17,  $P < \min ATC$  but  $P > \min AVC$ , the firm incurs loss, but can cover the fixed cost. They continue in the long run.

Consider a *perfectly competitive market with 1000 firms*. The cost schedule for each firm is given here:

1. If the price is BDT 12, What is the profit-maximizing/loss minimizing output?
2. Will each of the firms in the industry earn a profit or loss, when P=12?
3. How much is the individual profit/loss earned by the firms and how much is the industry profit/loss?

| Output (kg) | TC (BDT) | MC (BDT) | VC (BDT) | ATC (BDT) | AVC (BDT) |
|-------------|----------|----------|----------|-----------|-----------|
| 0           | 20       | -        | 0        | 0         | 0         |
| 1           | 40       | 20       | 20       | 40        | 20        |
| 2           | 55       | 15       | 35       | 27.5      | 17.5      |
| 3           | 67       | 12       | 47       | 22.33     | 15.67     |
| 4           | 82       | 15       | 62       | 20.5      | 15.5      |
| 5           | 102      | 20       | 82       | 20.4      | 16.4      |
| 6           | 128      | 26       | 108      | 21.33     | 18        |
| 7           | 160      | 32       | 140      | 22.85     | 20        |
| 8           | 197      | 37       | 177      | 24.63     | 22.135    |

## Solution to previous problem

1. At Price = 12, For Perfect competition,  
Profit is maximized where,  $MR=MC=12$ .  
Corresponding output is 3 kg (from given table)

2. Profit =  $TR - TC$

$$TR = 12 \times 3 = 36$$

$$TC = 67 \text{ (from table)}$$

$$36 - 67 = - 31 \text{ (loss)}$$

3. Individual loss = 31 tk

$$\text{Industry loss} = 31 \times 1000 = 31000 \text{ tk}$$

# Summary

- *Key concepts:*
  - Perfect competition (PC)
  - Profit maximizing output decisions
  - PC: Graphical representation
- *Reading: Chapter 11 of “Economics” by Parkin, Powell, Matthews 6th Ed.*